

# KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT

## GRADE 9 2025



### AGRICULTURE AND NUTRITION

Time: 2 hours

Name: .....

School: .....

Assessment Number: .....

Signature: ..... Date: .....

#### Instructions for Learners

1. Write your name, school name, and assessment number in the spaces provided.
2. This paper consists of **32 questions**.
3. Check to ensure all pages are printed and no questions are missing.
4. Answer all the questions in the spaces provided.
5. Sign at the end of this assessment.

#### SCORE GRID: SECTION A

| NO.     | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | TOTAL |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|-------|
| MAX     | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 20    |
| L.SCORE |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |       |

#### SCORE GRID: SECTION B

| NO.     | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | TOTAL |
|---------|----|----|----|----|----|----|----|----|----|----|----|----|-------|
| MAX     | 3  | 3  | 3  | 2  | 3  | 3  | 3  | 2  | 2  | 2  | 3  | 2  | 30    |
| L.SCORE |    |    |    |    |    |    |    |    |    |    |    |    |       |



## SECTION A. (20 MARKS)

**1.** Grade 9 learners are studying different methods of conserving forage and want to understand the one that involves cutting, drying and compressing forage into bales.

Which method involves cutting, drying and compressing forage into bales?

- A. Stacking
- B. Standing forage
- C. Baled hay making
- D. Box bailing

**2.** Bani owns a restaurant. She had some leftover pizza. How should she store it to maintain its quality?

- A. Wrap it in aluminum foil and leave it on the counter
- B. Store it in a cardboard box in the refrigerator
- C. Place it in a plastic bag and freeze
- D. Refrigerate it in an airtight container

**3.** Which one of the following organic gardening practices does not contribute to reducing pest problems naturally?

- A. Crop rotation
- B. Mulching
- C. Companion planting
- D. Using pesticide

**4.** A Grade 9 learner is baking bread and wants to use some leavening agents. Which one is the odd one out?

- A. Yeast
- B. Baking powder
- C. Butter
- D. Baking soda

**5.** During a routine check of the community storage unit, Meddy is tasked with ensuring the longevity of stored grains. What aspect should he regularly monitor?

- A. The colour of the storage unit's walls
- B. The moisture content of the grains
- C. The temperature outside the storage unit
- D. The height of the grain piles

**6.** Margy is a small-scale farmer using organic gardening practices. Which of the following crops did she not grow using organic gardening practices in her kitchen garden?

- A. Carrots
- B. Apples
- C. Basil
- D. Tomatoes

**7.** Larry noticed a bad smell coming from the kitchen sink. What should he do first to clean the sink properly at his household level?

- A. Clean the sink with appropriate disinfectants
- B. Paint the sink
- C. Replace the sink
- D. Ignore the smell

**8.** Amina harvested plenty of beans and her friend suggested the storage structure below for her. What is the name of the storage structure?



- A. Sacks
- B. Granary
- C. Boxes
- D. Sketch house

**9.** Jidena wanted to make chapatis for dinner for the first time. The following ingredients are needed for his cooking except:

- A. Milk
- B. All-purpose flour
- C. Cooking oil
- D. Water

**10.** A fisherman caught several fish from the fish pond at his farm. What is the first step he should do in preparing the fish?

- A. Washing
- B. Gutting
- C. Frying
- D. Scaling

**11.** Rita wants to cook her potatoes using dry heat and high temperatures for a crunchy result. Which method should she use?

- A. Boiling                      B. Drying
- C. Roasting                    D. Heating

**12.** After reading about pollution, Ella is motivated to keep her home's waste facilities clean. What is an immediate benefit of her action?

- A. Increased pollution
- B. More waste accumulation
- C. Better home hygiene
- D. Higher expenses

**13.** Nancy was putting plant remains and residues along the contours as shown below. What might be the reason for her to do so?



- A. To prevent weeds from growing
- B. To trap pests that may eat the crops
- C. To keep the soil warm
- D. To prevent moving soil materials from being swept away

**14.** The scientific term for the reaction between fats/oils and alkali that creates soap is:

- A. Saponification            B. Neutralization
- C. Combustion                D. Fermentation

**15.** Susan is researching what to avoid when disinfecting electronic devices. Which one is the best answer?

- A. Wiping with a dry cloth
- B. Using alcohol-based wipers

- C. Spraying directly with liquid
- D. Using a microfiber cloth

**16.** Jane and Doro have a bumper harvest of pumpkins. How can they add value to their pumpkins?

- A. Selling them fresh    B. Making pumpkin flour
- C. Letting them rot        D. Feeding them to livestock

**17.** Which of the following benefits does tree planting contribute towards environment conservation?

- A. Decrease water availability
- B. Increase plastic production
- C. Promote urbanization
- D. Reduce carbon dioxide level

**18.** Lily's mother always reminded her to wash fruits and vegetables before eating them. Why is washing fruits and vegetables important?

- A. To make them taste better
- B. To remove dirt, pesticide and bacteria
- C. To make them shiny
- D. To make them easier to peel

**19.** What benefit can one gain by enrolling in an online course that teaches eco-friendly techniques?

- A. Learning how to sell his farm
- B. Increasing his farm's experience
- C. Gaining knowledge about sustainable and eco-friendly farming practices
- D. Reducing the quality of his crops

**20.** Hadiya is a chef at Brightstar Academy. How can she ensure food safety during meal preparation?

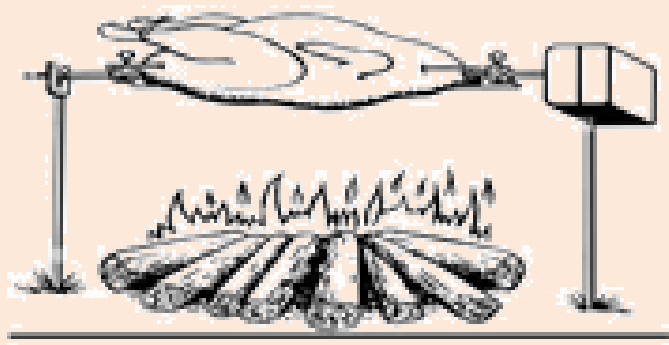
- A. Wash hands and surfaces frequently
- B. Use raw eggs in cooking
- C. Keep hot foods at room temperature
- D. Reheat food only once

SECTION B (30 Marks)

21. During a planting exercise, learners were instructed to mulch their potato seedlings and water them. Give **three benefits** they received from mulching their seedlings. (3 marks)

- i. ....
- ii. ....
- iii. ....

22. During Easter holiday, Bella prepared some food using the method shown below.



Identify **three precautionary measures** she was observing while preparing the meal. (3 marks)

- i. ....
- ii. ....
- iii. ....

23. The following is a warehouse a flour company has for its maize.



What are the **management practices** they carry out before milling maize? (3 marks)

- i. ....
- ii. ....
- iii. ....

24. State **two ways** of reducing surface runoff. (2 marks)

- i. ....
- ii. ....

25. Leila practices organic gardening on her farm. Outline **three organic practices** that she applies in her farm for crop production. (3 marks)

- i. ....
- ii. ....
- iii. ....

26. The picture below shows a crop management practice.



(a) What do we call the practice? (1 mark)

.....

(b) Identify **one tool** used to carry out the practice. (1 mark)

.....

27. Julia lives in a rural area where water scarcity is a major problem. To address this, they plan to implement a water storage method shown below to ensure a steady water supply throughout the year.

(a) Identify the water storage method above. (1 mark)

.....

(b) State **two uses** of harvested water. (2 marks)

- i. ....
- ii. ....

28. Kimtai decided to conserve his forage by using the stacking method.  
State **two advantages** of conserving forages using this method. (2 marks)

- i. ....
- ii. ....

29. A Grade 9 learner was asked by her teacher to name **two types of flour mixtures** used in food production.



What were her responses? (2 marks)

- i. ....
- ii. ....

30. Identify **two importance** of integrated farming. (2 marks)

- i. ....
- ii. ....

31. The picture below was taken by a Grade 9 learner from a famous poultry farmer during their visit to his farm.



(a) Identify the **poultry equipment** used by the farmer above. (1 mark)

.....

(b) State **two importance** of using animal waterers in animal rearing. (2 marks)

- i. ....
- ii. ....

32. Grace’s mother cooked food in excess for her birthday and she did not want to waste it, so she conserved it as leftover.

State **two importance** of conserving leftovers. (2 marks)

- i. ....
- ii. ....

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## SECTION A (20 MARKS)

(Each question = 1 mark)

| Qn | Answer                                                           | Explanation                                                     |
|----|------------------------------------------------------------------|-----------------------------------------------------------------|
| 1  | <b>C. Baled hay making</b>                                       | Forage is cut, dried, and compressed into bales.                |
| 2  | <b>D. Refrigerate it in an airtight container</b>                | Prevents contamination and maintains freshness.                 |
| 3  | <b>D. Using pesticide</b>                                        | Not a natural method of reducing pests.                         |
| 4  | <b>C. Butter</b>                                                 | Butter is not a leavening agent.                                |
| 5  | <b>B. The moisture content of the grains</b>                     | Moisture monitoring prevents spoilage.                          |
| 6  | <b>B. Apples</b>                                                 | Apples are less commonly grown in kitchen gardens organically.  |
| 7  | <b>A. Clean the sink with appropriate disinfectants</b>          | First step in removing odor-causing bacteria.                   |
| 8  | <b>B. Granary</b>                                                | Traditional storage structure for grains.                       |
| 9  | <b>A. Milk</b>                                                   | Chapati ingredients are flour, water, oil, and salt – not milk. |
| 10 | <b>B. Gutting</b>                                                | First step in fish preparation is removing internal organs.     |
| 11 | <b>C. Roasting</b>                                               | Uses dry heat at high temperature for crunchy results.          |
| 12 | <b>C. Better home hygiene</b>                                    | Keeping waste facilities clean prevents disease.                |
| 13 | <b>D. To prevent moving soil materials from being swept away</b> | Trash lines control soil erosion.                               |
| 14 | <b>A. Saponification</b>                                         | Reaction of fats/oils with alkali producing soap.               |

|    |                                                                                  |                                         |
|----|----------------------------------------------------------------------------------|-----------------------------------------|
| 15 | <b>C. Spraying directly with liquid</b>                                          | Damages electronics.                    |
| 16 | <b>B. Making pumpkin flour</b>                                                   | Value addition through processing.      |
| 17 | <b>D. Reduce carbon dioxide level</b>                                            | Trees absorb CO <sub>2</sub> .          |
| 18 | <b>B. To remove dirt, pesticide and bacteria</b>                                 | Ensures food safety.                    |
| 19 | <b>C. Gaining knowledge about sustainable and eco-friendly farming practices</b> | Benefit of online eco-friendly courses. |
| 20 | <b>A. Wash hands and surfaces frequently</b>                                     | Key step in food safety.                |

## SECTION B – MARKING SCHEME (30 Marks)

**21.** Benefits of mulching potato seedlings: (Any 3 × 1 = 3 marks)

- i. Conserves soil moisture by reducing evaporation.
- ii. Prevents weed growth.
- iii. Reduces soil erosion.
- iv. Improves soil fertility as mulch decomposes.
- v. Regulates soil temperature.

**22.** Precautionary measures Bella was observing while preparing food: (Any 3 × 1 = 3 marks)

- i. Washing hands and cooking utensils before food preparation.
- ii. Keeping raw and cooked food separate to avoid contamination.
- iii. Cooking food thoroughly to kill germs.
- iv. Covering food while cooking to prevent dirt or insects from entering.
- v. Using clean water and fresh ingredients.

**23.** Management practices before milling maize: (Any 3 × 1 = 3 marks)

- i. Sorting maize to remove stones, sticks, and foreign matter.
- ii. Drying maize to reduce moisture content.
- iii. Cleaning maize to remove dust and dirt.
- iv. Storing maize properly to prevent pest attack.

**24.** Ways of reducing surface runoff: (Any 2 × 1 = 2 marks)

- i. Planting cover crops.



- ii. Constructing terraces.
- iii. Mulching.
- iv. Planting grass strips.
- v. Building contour bunds.

**25.** Organic gardening practices applied by Leila: (Any  $3 \times 1 = 3$  marks)

- i. Crop rotation.
- ii. Use of organic manure/compost.
- iii. Mulching.
- iv. Companion planting.
- v. Biological pest control.

**26.** (a) Name of the practice: (1 mark)

– **Weeding**

(b) Tool used: (Any  $1 \times 1 = 1$  mark)

– Hoe / Hand fork / Panga

**27.** (a) Water storage method: (1 mark)

– **Water tank / Water reservoir / Dam**

(b) Uses of harvested water: (Any  $2 \times 1 = 2$  marks)

- i. Irrigation of crops.
- ii. Domestic use (washing, cooking, cleaning).
- iii. Drinking water for animals.
- iv. Construction or farm activities.

**28.** Advantages of conserving forage using stacking: (Any  $2 \times 1 = 2$  marks)

- i. Forage is protected from spoilage.
- ii. Saves storage space.
- iii. Makes handling and transportation easier.
- iv. Maintains feed availability during dry season.

**29.** Types of flour mixtures: (Any  $2 \times 1 = 2$  marks)

- i. Stiff dough (e.g., chapati, mandazi).
- ii. Pour batters (e.g., pancakes).
- iii. Drop batters (e.g., muffins).
- iv. Soft dough (e.g., bread).

**30.** Importance of integrated farming: (Any  $2 \times 1 = 2$  marks)

- i. Maximizes use of farm resources.

- ii. Provides both food and income.
- iii. Reduces risk of total loss.
- iv. Improves soil fertility through recycling of waste.
- v. Enhances biodiversity.

**31.** (a) Poultry equipment: (1 mark)

– **Waterer / Drinker**

(b) Importance of using animal waterers: (Any 2 × 1 = 2 marks)

- i. Provides clean and safe water for animals.
- ii. Prevents wastage of water.
- iii. Reduces contamination of water.
- iv. Saves farmer's time and effort.

**32.** Importance of conserving leftovers: (Any 2 × 1 = 2 marks)

- i. Reduces food wastage.
- ii. Saves money.
- iii. Provides food for later use.
- iv. Promotes food security in the household.

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